Comparison, Initial-Layer Positivity, and Reflection Symmetry for the Impulsively Started Cylinder

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Abstract

We establish conditional a priori estimates for a given solution of the cylinder Prandtl equations with impulsive datum $u(0, R, \theta) = 2\sin\theta$. The sole quantitative tangential regularity assumption in the comparison estimates is $u_\theta \in L^1_t L^\infty_{R,\theta}$. The solution is assumed classical at positive times and to have the natural energy trace for its defect from the outer velocity. We prove a uniform sine envelope, a Bernstein estimate of order $t^{-1/2}$ for the normal derivative, and strict positivity for a universal short time. The positivity proof works directly with the given impulsive solution: it uses a Gaussian defect bound and a strong negative-part trace for the normal derivative, rather than an approximation by solutions with potentially different lifespans. Reflection symmetry is treated separately under a fixed tangential analyticity radius and an integrable normal-moment bound, by a direct Osgood uniqueness argument. These statements do not establish existence of the impulsive solution or prevent later boundary-layer separation.

1 Setting, hypotheses, and conclusions

The physical variables of the leading boundary layer are the angle θ and the stretched normal distance $R = (r-1)/\varepsilon$. Write

$$\mathcal{O} = (0,\infty)_R \times (0,\pi)_\theta, \qquad V(\theta) = 2\sin\theta, \qquad V' = 2\cos\theta.$$

The argument order of a velocity is (t, R, θ). On the upper semicircle consider

$$u_t + uu_\theta + vu_R - u_{RR} = VV' = 2\sin(2\theta), \tag{1.1}$$

$$u_\theta + v_R = 0, \qquad v(t,R,\theta) = -\int_0^R u_\theta(t,Z,\theta)\,dZ. \tag{1.2}$$

The wall, axis, matching, and initial conditions are

$$u(t,0,\theta) = v(t,0,\theta) = 0, \qquad u(t,R,0) = u(t,R,\pi) = 0, \tag{1.3}$$

$$u(t,R,\theta) \longrightarrow V(\theta) \quad (R\to\infty), \qquad u(0,R,\theta) = V(\theta) \quad (R>0). \tag{1.4}$$

The initial condition is not a continuous corner condition at $t = R = 0$.

Fix an interval $(0, T]$. Our solution hypotheses are as follows.

1. The equations and boundary traces hold classically for positive times, with sufficient local differentiability to differentiate once in R and apply the chain rules used below. Smoothness up to the finite boundaries on every compact positive-time interval is sufficient. The corresponding computations may also be used for weak solutions for which these chain rules, traces, and truncation inequalities are justified independently.

2. With $w = u - V$, the natural energy and initial-trace conditions hold:

$$w \in C([0,T]; L^2(\mathcal{O})), \qquad w(0) = 0, \qquad u_R \in L^2((0,T) \times \mathcal{O}). \tag{1.5}$$

In particular, it is $u - V$, not u, that lies in L^2 on the half-line.

3. The tangential derivative satisfies

$$k(t) := \|u_\theta(t)\|_{L^\infty(\mathcal{O})} \in L^1(0,T), \qquad J(t) = \int_0^t k(s)\, ds. \tag{1.6}$$

The outer matching is understood uniformly in θ at regular times; alternatively one may use the energy truncations in the proof. By (1.2),

$$|v(t,R,\theta)| \le k(t)R. \tag{1.7}$$

We do not assert that every distributional solution satisfying only (1.5)–(1.6) possesses the differentiability in item 1. Establishing such regularization or the existence of this solution is a separate problem.

Theorem 1.1 (Comparison bounds)**.** *Under the hypotheses above, for $0 < t \le T$ and $0 \le \theta \le \pi$,*

$$-V(\theta)\tanh(2t) \le u(t,R,\theta) \le V(\theta). \tag{1.8}$$

Consequently $|u| \le 2\sin\theta$. If $\sigma(t) = \min\{t, 1/8\}$, then

$$|u_R(t,R,\theta)| \le V(\theta)\sqrt{\sigma(t)^{-1} + 13}. \tag{1.9}$$

Moreover, for $0 < t \le \min\{T, 10^{-4}\}$,

$$\boxed{u(t,R,\theta) \ge V(\theta)\left(1 - \exp\left[-\frac{R}{1200\sqrt{t}}\right]\right).} \tag{1.10}$$

Thus $u > 0$ for $R > 0$ and $0 < \theta < \pi$ on this interval. The constants are deliberately nonoptimal. In particular, 10^{-4} is a proved sufficient time, not an assertion about the optimal onset time of reverse flow.

2 Scalar comparison on the unbounded strip

Set

$$\mathcal{L} = \partial_t + u\partial_\theta + v\partial_R - \partial_{RR}.$$

The transport vector (u, v) is divergence free in (θ, R) and is tangent to the finite boundaries in (1.3). The usual positive-part energy argument therefore applies to scalar inequalities

$$\mathcal{L}z + cz \le 0, \qquad c^- \in L^1_t L^\infty,$$

when the positive initial and boundary traces vanish and the positive part has the indicated energy integrability. Here and below, all arguments on the half-line are made first with cutoffs.

For clarity, a useful cutoff is

$$\chi_L(t,R) = \chi\left(\frac{Re^{J(t)}}{L}\right),$$

where χ is smooth, decreasing, equal to one on $[0,1]$, and zero on $[2,\infty)$. By (1.7),

$$(\partial_t + v\partial_R)\chi_L = \chi'\left(\frac{Re^J}{L}\right)\frac{e^J}{L}(kR + v) \le 0. \tag{2.1}$$

Its diffusion error is bounded by $Ce^{2J(T)}L^{-2}$. Energy integrability makes the resulting error tend to zero. Convex smooth approximations to the positive part, followed by monotone or dominated convergence, justify the truncation identities. Thus the linearly growing normal drift creates no boundary contribution at infinity.

For the upper bound, $z = u - V$ satisfies exactly

$$\mathcal{L}z + V'z = 0.$$

Its initial L^2 trace is zero, its wall trace is $-V \le 0$, and its axis traces vanish. Since $|V'| \le 2$, comparison gives $u \le V$.

For the lower bound, let $C(t) = 2\tanh(2t)$ and $h = u + C(t)\sin\theta$. Using $C' = 4 - C^2$, direct calculation gives

$$\mathcal{L}h - C\cos\theta\, h = (4 - C^2)\sin\theta(1 + \cos\theta) \ge 0. \tag{2.2}$$

Its initial trace is $V \ge 0$ and its finite boundary traces are nonnegative. Moreover $h^- \le |u - V|$, so the same energy comparison is admissible. This proves (1.8).

3 A Bernstein normal-derivative estimate

For $0 < \theta < \pi$ define

$$q = \frac{u}{V}, \qquad W = q_R = \frac{u_R}{V}.$$

The envelope gives $-1 \le q \le 1$. The quotient is used only inside the open semicircle; its weighted estimates below remain regular at the axes. Direct calculation, using $v_R = -u_\theta$, yields

$$\mathcal{L}q = V'(1 - q^2), \qquad \mathcal{L}W = -V'qW. \tag{3.1}$$

The cancellation in the second equation is essential. The wall traces imply

$$q(t, 0, \theta) = 0, \qquad W_R(t, 0, \theta) = -V'(\theta). \tag{3.2}$$

Fix a positive starting time a and put $s = t - a$, with $0 \le s \le 1/8$. Consider

$$B = sW^2 + q^2 + 2sV'q.$$

The last term corrects the inhomogeneous normal derivative at the wall. Indeed $B_R|_{R=0} = 0$. Since $\mathcal{L}V' = -V^2 q$, (3.1) gives

$$\begin{aligned}\mathcal{L}B = {} & (-1 - 2sV'q)W^2 - 2sW_R^2 \\ & + 2V'q(1 - q^2) + 2V'q + 2s\{(V')^2(1 - q^2) - V^2q^2\} \le 9.\end{aligned} \tag{3.3}$$

Here $|V|, |V'| \le 2$, $|q| \le 1$, and $s \le 1/8$ suffice. At $t = a$, $B = q^2 \le 1$.

We record why this comparison does not assume a bounded initial normal derivative. Let

$$z = (B - 1 - 9s)_+.$$

It satisfies $\mathcal{L}z \le 0$ in the convex-truncation sense and $z_R|_{R=0} = 0$. Furthermore

$$V^2 z \le s u_R^2, \tag{3.4}$$

since $q^2 \le 1$ and $2sV'q \le 4s$. Multiply the inequality by $V^2\chi_L$ and integrate. The only nonzero weight-transport term obeys

$$\left| u \frac{\partial_\theta(V^2)}{V^2} \right| = |2qV'| \le 4.$$

One may first cut off the two axes; the corresponding errors tend to zero by (3.4) and $|u| \le V$. The normal cutoff is handled by (2.1). At the wall the normal derivative vanishes. Letting the cutoffs tend to their limits gives

$$M_a'(t) \le 4M_a(t), \qquad M_a(t) := \int_{\mathcal{O}} V^2 z(t).$$

The initial trace for this integral deserves attention: pointwise $B(a) = q(a)^2$ alone does not justify a global integral trace. Instead, (3.4) gives

$$0 \le M_a(a+h) \le h \, \|u_R(a+h)\|_2^2 .$$

Local time integrability of $\|u_R\|_2^2$ provides a sequence $h_n \downarrow 0$ with $h_n \, \|u_R(a+h_n)\|_2^2 \to 0$; otherwise its integral would dominate a multiple of $\int_0 dh/h$. Apply Gronwall from $a + h_n$ and let $n \to \infty$. Thus $M_a(t) = 0$ and $B \le 1 + 9s$. Since $B \ge sW^2 - 4s$,

$$W^2 \le s^{-1} + 13.$$

For $t \le 1/8$, let $a \downarrow 0$ at fixed t. For $t > 1/8$, start at $a = t - 1/8$. This proves (1.9). No bound on u_R at $t = 0$ has been used.

4 A Gaussian defect bound and the initial derivative trace

The positivity proof needs a genuine initial trace for a derivative comparison. A positive distributional measure trace alone would not suffice for the negative-part estimate below.

4.1 Gaussian decay of the velocity defect

Define

$$\tau(t) = \int_0^t e^{-2J(s)} \, ds, \qquad P(t,R) = e^{4t} \operatorname{erfc}\left(\frac{e^{-J(t)} R}{2\sqrt{\tau(t)}} \right).$$

Here $\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}} \int_z^\infty e^{-\zeta^2} \, d\zeta$. We have $P_R \le 0$ and

$$(\partial_t + k(t) R \partial_R - \partial_{RR}) P = 4P.$$

Thus, by $v \le kR$,

$$(\partial_t + v\partial_R - \partial_{RR}) P \ge 4P.$$

For $w = V - u \ge 0$, $\mathcal{L}w + V'w = 0$. Moreover

$$(\mathcal{L} + V')(VP) = V\{(\partial_t + v\partial_R - \partial_{RR})P + V'(1+q)P\} \ge 0.$$

The initial defect trace is zero and $VP|_{R=0} = Ve^{4t} \ge V = w|_{R=0}$. Comparison gives

$$0 \le V - u \le VP, \qquad 0 \le 1 - q \le C \exp(-cR^2/t). \tag{4.1}$$

The constants in the last estimate may depend on T and $J(T)$; they will be used only to justify limits. In particular,

$$\|V - u\|_{L^1(\mathcal{O})} \le \frac{8e^{4t+J(t)}}{\sqrt{\pi}} \sqrt{\tau(t)} \le C\sqrt{t}. \tag{4.2}$$

We used $\int_0^\pi V \, d\theta = 4$ and $\int_0^\infty \operatorname{erfc} z \, dz = 1/\sqrt{\pi}$.

4.2 Comparison with the heat initial layer

Let

$$Q(t,R)=\operatorname{erf}\left(\frac{R}{2\sqrt{t}}\right),\qquad E=u-VQ,\qquad A(t)=t+J(t).$$

Then E has zero initial L^2 trace and homogeneous wall and axis traces, and

$$\mathcal{L}E=f:=VV'(1-qQ)-vVQ_R. \tag{4.3}$$

The Gaussian estimate, $|v|\le kR$, and the explicit heat kernel give

$$\|f(t)\|_2\le Ct^{1/4}(1+k(t)),\qquad \|Rf(t)\|_2\le Ct^{3/4}(1+k(t)). \tag{4.4}$$

Here and below all norms in this section are over $\mathcal{O}$. Testing (4.3) against E gives

$$\|E(t)\|_2\le Ct^{1/4}A(t),\qquad \int_0^t\|E_R(s)\|_2^2\,ds\le C\sqrt{t}\,A(t)^2. \tag{4.5}$$

For example, $\|E(t)\|_2\le\int_0^t\|f(s)\|_2\,ds$, and the integrated dissipation is bounded by the product of this integral and $\sup_{s\le t}\|E(s)\|_2$.

To justify a normal moment without assuming one in the initial datum, introduce bounded weights

$$m_L(R)=\frac{R^2}{1+R^2/L^2},\quad 0\le m_L\uparrow R^2,\quad |Rm_L'|\le 2m_L,\quad m_L''\le 2.$$

Set $Z_L=\int m_LE^2$ and $D_L=\int m_LE_R^2$. The initial value of Z_L is zero because m_L is bounded. Integration by parts gives

$$\frac12 Z_L'+D_L\le\|Rf\|_2\sqrt{Z_L}+kZ_L+\|E\|_2^2.$$

Gronwall and the elementary inequality $ab\le a^2/2+b^2/2$, applied to the time-integrated inequality and its running supremum, give

$$Z_L(t)+\int_0^t D_L(s)\,ds\le C\left[\left(\int_0^t\|Rf(s)\|_2\,ds\right)^2+\int_0^t\|E(s)\|_2^2\,ds\right]\le Ct^{3/2}A(t)^2,$$

uniformly in L. Monotone convergence yields

$$\|RE(t)\|_2^2+\int_0^t\|RE_R(s)\|_2^2\,ds\le Ct^{3/2}A(t)^2. \tag{4.6}$$

Averaging (4.5) and (4.6) on intervals $(t/2,t)$ produces a sequence $t_n\downarrow 0$ such that

$$\sqrt{t_n}\left(\|E_R(t_n)\|_2^2+t_n^{-1}\|RE_R(t_n)\|_2^2\right)\longrightarrow 0.$$

Weighted Cauchy–Schwarz gives

$$\|E_R(t_n)\|_1^2\le\frac{\pi^2}{2}\sqrt{t_n}\left(\|E_R(t_n)\|_2^2+t_n^{-1}\|RE_R(t_n)\|_2^2\right)\longrightarrow 0.$$

Since $VQ_R\ge 0$, we have proved the needed strong trace:

$$\boxed{\left\|(u_R)^-(t_n)\right\|_{L^1(\mathcal{O})}\longrightarrow 0.} \tag{4.7}$$

This concerns a sequence of times of the same given solution; no approximate solution or common lifespan is involved.

5 A derivative barrier and strict short-time positivity

Set $T_* = \min\{T, 10^{-4}\}$ and fix $0 < d \le 5 \cdot 10^{-5}$. Define

$$h(q) = \frac{1}{400}(1-q)\left(1 - \frac{1}{2}e^{-256q}\right), \qquad H_d(t,q) = \frac{h(q)}{\sqrt{t+d}}, \qquad G_d = u_R - VH_d(t,q).$$

By (1.8), for $0 < t \le T_*$,

$$-\frac{1}{1024} \le q \le 1.$$

On this interval elementary differentiation gives

$$\frac{1-q}{1200} \le h(q) \le \frac{1-q}{400}, \qquad h''(q) < 0, \tag{5.1}$$

$$|h(q)h''(q)| \le 0.104, \qquad |(1-q^2)h'(q) + qh(q)| \le 511h(q), \tag{5.2}$$

$$h(0)h'(0) = \frac{255}{640000}, \qquad h'(0) > 0. \tag{5.3}$$

For completeness, put $z = e^{-256q}$ and $r = 1 - q \le 1025/1024$. Since $z \le e^{1/4} < 4/3$, the bounds in (5.1) follow. Also

$$|hh''| = \frac{128}{160000} r(2+256r)z(1-z/2) \le \frac{64}{160000}\frac{1025}{1024}\left(2 + 256\frac{1025}{1024}\right) < 0.104.$$

For $q < 1$,

$$\frac{(1-q^2)h' + qh}{h} = -1 + \frac{128e^{-256q}(1-q^2)}{1 - \frac{1}{2}e^{-256q}},$$

and $e^{1/4} < 1.3$ bounds its absolute value by 511; the endpoint follows by continuity.

The physical normal derivative $\omega = u_R$ satisfies $\mathcal{L}\omega = 0$. Consequently

$$\mathcal{L}G_d - c_d G_d = -V\mathcal{R}_d, \quad c_d = (H_d)_{qq}(W + H_d), \tag{5.4}$$

where

$$\mathcal{R}_d = (H_d)_t + V'\{(1-q^2)(H_d)_q + qH_d\} - H_d^2(H_d)_{qq}.$$

For $q < 1$, (5.2) implies

$$\frac{\mathcal{R}_d}{H_d} \le \frac{-1/2 + 0.104}{t+d} + 1022 < 0,$$

because $t + d \le 1.5 \cdot 10^{-4}$. At $q = 1$, $\mathcal{R}_d = 0$. Thus $\mathcal{L}G_d - c_d G_d \ge 0$.

At the wall, $u_{RR} = -VV'$ and $q = 0$, hence

$$(G_d)_R + b_d G_d = -V\{V' + H_d(t,0)(H_d)_q(t,0)\} \le 0, \qquad b_d = (H_d)_q(t,0) > 0. \tag{5.5}$$

Indeed,

$$H_d(t,0)(H_d)_q(t,0) = \frac{255}{640000(t+d)} \ge 2.65625 > 2 \ge |V'|.$$

For each fixed d, the Bernstein estimate implies

$$\|c_d(t)\|_\infty \le C_d(1 + t^{-1/2}) \in L^1(0, T_*), \qquad \sup_{0<t<T_*} b_d(t) < \infty. \tag{5.6}$$

It remains to justify the initial comparison. From (5.1) and (4.2),

$$0 \le VH_d \le \frac{V-u}{400\sqrt{d}}, \qquad \|VH_d(t,q)\|_1 \le C_d\sqrt{t}.$$

Together with (4.7), this yields

$$\|(G_d)^-(t_n)\|_1 \longrightarrow 0 \quad \text{for each fixed } d > 0. \tag{5.7}$$

Here is a weighted L^1 comparison accommodating the Robin inequality (5.5). Put $z = (G_d)^-$, choose $B > 2 + \sup b_d$, and let $\psi(R) = (1+R)^{-B}$. Convex truncation gives $\mathcal{L}z \le c_d^+ z$ and, at the wall, $z_R + b_d z \ge 0$. For $M(t) = \int_{\mathcal{O}} \psi z$, integration by parts gives

$$M' \le \{\|c_d\|_\infty + Bk + B(B+1)\}M.$$

The wall contribution is

$$-z_R(t,0,\theta) + \psi'(0)z(t,0,\theta) \le (b_d - B)z(t,0,\theta) \le 0.$$

At infinity the cutoff errors vanish: (1.9) bounds u_R for positive times and ψ is integrable, while $|v| \le kR$. The axis fluxes vanish. Start the inequality at t_n, use (5.7) and (5.6), and let $n \to \infty$. Gronwall gives $M(t) = 0$, hence $G_d \ge 0$.

Only now let $d \downarrow 0$. The solution has never changed. Using (5.1),

$$q_R \ge \frac{h(q)}{\sqrt{t}} \ge \frac{1-q}{1200\sqrt{t}}.$$

Since $q(t,0,\theta) = 0$, integration in R gives (1.10). It also gives $u_R \ge 0$ on this short interval. At the wall,

$$\frac{V(\theta)}{1200\sqrt{t}} \le u_R(t,0,\theta) \le V(\theta)\sqrt{t^{-1}+13}.$$

Thus the $t^{-1/2}$ initial-layer order is sharp at every fixed interior angle. This is an initial-boundary incompatibility effect, not a finite positive-time blowup theorem.

6 Reflection symmetry and its regularity requirements

Work now on the full periodic domain $\mathbb{R}_+ \times \mathbb{T}_{2\pi}$. The reflection transform is

$$\widetilde{u}(t,R,\theta) = -u(t,R,-\theta), \qquad \widetilde{v}(t,R,\theta) = v(t,R,-\theta). \tag{6.1}$$

The forcing $2\sin(2\theta)$ and outer datum $2\sin\theta$ are odd, while the normal datum $-2R\cos\theta$ is even.

Theorem 6.1 (Reflection symmetry under explicit analytic bounds)**.** *Let $\mathcal{H} = \mathbb{R}_+ \times \mathbb{T}_{2\pi}$. Suppose a positive-time classical solution satisfies the periodic equations, wall and matching conditions, the impulsive defect trace $u - V \in C([0,T]; L^2(\mathcal{H}))$, $(u-V)(0) = 0$, and $u_R \in L^2((0,T)\times\mathcal{H})$. Assume $k = \|u_\theta\|_\infty \in L^1(0,T)$ and, additionally, for some fixed $0 < \rho \le 1$,*

$$\begin{aligned} &\sup_{0<t\le T} \left\| e^{\rho|D_\theta|}(u-V)(t) \right\|_2 \le A < \infty, \\ &B(t) := \operatorname*{ess\,sup}_\theta \left(\int_0^\infty R|u_R(t,R,\theta)|^2\, dR \right)^{1/2} \in L^1(0,T). \end{aligned} \tag{6.2}$$

Here $e^{\rho|D_\theta|}$ multiplies the nth periodic Fourier coefficient by $e^{\rho|n|}$. Then

$$u(t,R,\theta) = -u(t,R,-\theta), \qquad v(t,R,\theta) = v(t,R,-\theta). \tag{6.3}$$

In particular, periodicity gives $u(t,R,0) = u(t,R,\pi) = 0$, and differentiability gives $v_\theta = 0$ on these axes.

Proof. We first prove uniqueness within the stated class. Let (u_1, v_1) and (u_2, v_2) be two solutions with the same data, satisfying the hypotheses with common bounds for the fixed analytic radius and norm. Write

$$w = u_1 - u_2, \qquad W_0 = v_1 - v_2 = -\int_0^R w_\theta(t, Z, \theta)\, dZ.$$

The exact difference equation is

$$w_t + u_1 w_\theta + v_1 w_R - w_{RR} = -(u_2)_\theta w - W_0 (u_2)_R. \tag{6.4}$$

For each θ, Cauchy–Schwarz gives

$$|W_0(R, \theta)|^2 \le R \int_0^\infty |w_\theta(Z, \theta)|^2\, dZ.$$

Consequently, with B_2 defined by (6.2) for u_2,

$$\|W_0 (u_2)_R\|_2 \le B_2(t)\, \|w_\theta\|_2 .$$

The divergence-free energy identity, justified by cutoffs and $|v_1| \le k_1 R$, gives, for $E(t) = \|w(t)\|_2^2$,

$$\frac{1}{2} E' + \|w_R\|_2^2 \le k_2 E + B_2\, \|w_\theta\|_2 \sqrt{E}. \tag{6.5}$$

We supply the analytic interpolation that controls the derivative loss. Put $F = \|w\|_2$ and $M = 1 + 2A$. If $F > 0$, split the Fourier series at

$$N = 1 + \frac{2}{\rho} \log \frac{M}{F}.$$

The low frequencies contribute at most NF to $\|w_\theta\|_2$. For the high frequencies,

$$\begin{aligned} \left\| \mathbf{1}_{|D_\theta| > N} w_\theta \right\|_2 &\le \frac{2}{e\rho} e^{-\rho N/2} \left\| e^{\rho |D_\theta|} w \right\|_2 \\ &\le \frac{2}{e\rho} F, \end{aligned}$$

using $\sup_{\xi \ge 0} \xi^2 e^{-\rho \xi} = 4/(e^2 \rho^2)$ and $\left\| e^{\rho |D_\theta|} w \right\|_2 \le 2A < M$. Hence

$$\|w_\theta\|_2 \le \frac{C}{\rho} F \left(1 + \log \frac{M}{F} \right). \tag{6.6}$$

The inequality is interpreted by continuity at $F = 0$. Substitution into (6.5) yields

$$E' \le C \left(k_2 + \frac{B_2}{\rho} \right) E \left(1 + \log \frac{M^2}{E} \right).$$

The coefficient is integrable. The function $z(1 - \log z)$ on $0 < z \le 1$ is an Osgood modulus because

$$\int_{0^+} \frac{dz}{z(1 - \log z)} = \infty.$$

To see the conclusion directly, integrate the separated differential inequality from a positive starting time, and then let that time tend to zero using $E(0) = 0$. A positive value at a later time would require a finite upper bound for the displayed divergent integral. Thus $E \equiv 0$ and the two solutions coincide.

Finally, direct substitution shows that the reflected pair (6.1) satisfies the same equations and data. In fact,

$$\widetilde{u}_\theta = u_\theta(t, R, -\theta), \quad \widetilde{u}_R = -u_R(t, R, -\theta), \quad \widetilde{v}_R = v_R(t, R, -\theta),$$

so incompressibility is preserved, while each term in the tangential equation becomes the negative of its value at $-\theta$. The forcing is odd, so the transformed equation has the original forcing. All norms in the hypotheses are preserved by reflection. Uniqueness therefore proves (6.3). □

The normal-moment condition in (6.2) admits the impulsive heat layer. For

$$u_R = \frac{V(\theta)}{\sqrt{\pi t}} e^{-R^2/(4t)}, \qquad B(t) = \frac{\|V\|_\infty}{\sqrt{\pi}}.$$

Thus this theorem does not impose bounded initial u_R. It does impose explicit additional analytic and normal-moment bounds; they have not been deduced from $u_\theta \in L^1_t L^\infty$ alone.

Tangential analytic well-posedness is also the framework of [1]. Its published initial-data assumptions include no-slip compatibility, so it cannot simply be invoked at the impulsive corner. The direct argument above avoids that issue. The scalar maximum principle alone does not prove full nonlinear uniqueness: the term $W_0(u_2)_R$ in (6.4) contains a tangential derivative of the difference. Analytic interpolation and Osgood's argument are what close this estimate.

Remark 6.2 (Axis identities versus full reflection symmetry). The tangential axis conditions can also be proved separately when the necessary traces exist. At a stagnation axis $\theta_0 \in \{0, \pi\}$ put $a(t, R) = u(t, R, \theta_0)$, $b(t, R) = u_\theta(t, R, \theta_0)$, and $V_0(t, R) = v(t, R, \theta_0)$. Then

$$a_t + V_0 a_R - a_{RR} + ba = 0, \qquad (V_0)_R = -b.$$

With zero initial and wall traces, the energy identity is

$$\frac{1}{2}\frac{d}{dt}\|a\|_2^2 + \|a_R\|_2^2 + \frac{3}{2}\int_0^\infty ba^2\, dR = 0.$$

An integrable L^∞ bound for b gives $a \equiv 0$ by Gronwall. This restricted argument proves the axis values, not the full odd symmetry. Evenness of v, and hence $v_\theta = 0$ on the axes when that trace exists, follows from the full reflection statement.

References

[1] I. Kukavica and V. Vicol, *On the local existence of analytic solutions to the Prandtl boundary layer equations*, Communications in Mathematical Sciences **11** (2013), no. 1, 269–292. doi:10.4310/CMS.2013.v11.n1.a8.